JACOB PILAND

Computer Vision & Trustworthy AI Researcher | Ph.D. Computer Science
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PROFESSIONAL SUMMARY

Computer vision and trustworthy AI researcher, with expertise in synthetic media detection, multimodal vision-language modeling, salience-guided learning, energy-based training, and robust visual explanation. Develops applied computer-vision methods for understanding generated imagery, improving open-set generalization, and evaluating model behavior across large-scale image benchmarks. Research spans synthetic face detection across GAN families, multimodal LLM prompting for visual biometric inference, CAM-based model-focus control, adversarially robust saliency methods, and statistically rigorous evaluation pipelines. Strong engineering background in Python, PyTorch, C/C++, reproducible research systems, and cross-disciplinary technical collaboration.

CORE COMPETENCIES & TECHNICAL SKILLS

Computer Vision / Generative AI	Synthetic image detection; multimodal vision-language models; salience visualization; GradCAM/CAM methods; open-set generalization; GAN-generated imagery; biometric image analysis
ML / Modeling	PyTorch; DenseNet; ResNet; Inception; ConvNeXt; energy-based models; calibration; adversarial robustness; statistical evaluation
Engineering Strengths	Model evaluation; calibration analysis; adversarial robustness testing; open-set benchmarking; Python; C/C++; R; reproducible research workflows

EDUCATION

Ph.D., Computer Science | University of Notre Dame du Lac — May 2026

M.S., Computer Science | University of Notre Dame du Lac — May 2022

B.S., Computer Science; B.S., Biochemistry | Utah State University — May 2018

RESEARCH EXPERIENCE

Doctoral Candidate / Researcher | University of Notre Dame | *South Bend, IN* — **2021-2026**

- Designed and ran a 16-condition comparative study across Gemini 2.5 Pro and locally hosted Llama 3.2-Vision:90b, evaluating structured prompting and human or machine-expanded salience for binary iris presentation attack detection. Gemini 2.5 Pro + structured prompt + human salience achieved 0.053 MSE, outperforming both human examiners (0.062) and a salience-guided CNN baseline (0.345 ± 0.041).
- Created a Machine-Expanded Saliency from Human pipeline that uses LLMs to convert sparse human observations into comprehensive image descriptions for few-shot biometric prompting.
- Showed that pre-trained vision-language representations encode useful biometric attack structure by demonstrating partial attack-type clustering in vision embeddings and improved live-vs.-spoof separation after minimal semantic guidance.
- Designed a 16-condition comparative study across Gemini 2.5 Pro and locally hosted Llama 3.2-Vision:90b, evaluating structured prompting and human or machine-expanded salience for binary iris PAD.
- Benchmarked synthetic-face detectors across training, validation, and open-set test splits that included FRGC, CelebA-HQ, FFHQ, SREFI, ProGAN, StarGANv2, StyleGAN, StyleGAN2, StyleGAN3, and StyleGAN2-ADA.
- Developed REBIS, a three-component loss function integrating classification loss, low-entropy saliency control, and energy-based regularization, enabling complementary optimization at both the sample level and dataset level to improve accuracy, calibration, and adversarial robustness simultaneously in synthetic face detection; optimal configuration achieved 0.738 AUROC, 0.935

Inverted ECE, and PGD robustness of ϵ = 2.232 against baselines of 0.563, 0.906, and 0.274 respectively.
- Developed DROID, a low-CAM-entropy training objective for synthetic face detection; best hybrid configuration achieved 0.783 AUROC versus 0.561 for cross-entropy and 0.636 for prior saliency-guided training, with only ~10% added training overhead.
- Proposed NG-EBM, a non-generative energy-based training strategy that eliminates SGLD sampling; achieved a 6× reduction in training cost relative to JEM (15 vs. 85 GPU hours per model) while reducing Expected Calibration Error from 5.0% to 1.8% on CIFAR10 and 22.7% to 3.3% on CIFAR100.
- Built and executed a cross-domain robustness evaluation pipeline spanning few-class iris PAD and 1,000-class ImageNet, testing saliency fidelity and susceptibility across DenseNet-121, ResNet-50, Inception-v3, and ConvNeXt-Tiny.
- Designed contrastive saliency-guided training methods (Per-class Salience, Contrast Salience) for binary classification that supervise both true- and false-class CAM behavior, improving out-set recall across medical imaging, biometric PAD, and synthetic face detection.
- Released source code, model weights, and training configurations for multiple computer-vision research projects, supporting reproducible experimentation and downstream model reuse.

ADDITIONAL ENGINEERING & RESEARCH EXPERIENCE

Research Assistant | University of Notre Dame | *South Bend, IN* **2018-2021**
- Developed network science-based analytical methods for open-ended protein-folding studies without ground-truth labels, enabling structural analysis under underdetermined conditions.
- Modeled protein structures as graphs to apply network-theoretic metrics and topology-driven analysis to biological systems.

Research Assistant | Utah State University | *Logan, UT* **2014 - 2018**
- Authored large-scale cluster analysis of cognitive effects of computer-based scaffolding across U.S. states and international cohorts under an NSF REESE-funded project.

Programming consultant | NASA Earth Science (Armstrong / UC Irvine) | *Irvine, CA* **Summer 2018**
- Advised software development across 28 research intern projects spanning C++, Python, MATLAB, and R; delivered weekly programming instruction for interdisciplinary science teams.

Software Engineering Intern | Hewlett Packard Enterprise | *Boise, ID* **Summer 2017**
- Improved low-level C driver quality and maintainability by increasing power-failure code coverage 5x, improving runtime by approximately 30%, and reducing legacy code size by approximately 30% through refactoring.

Research Intern | NASA Earth Science (Armstrong / UC Irvine) | *Irvine, CA* **Summer 2015**
- Modeled atmospheric ozone production in the Los Angeles Basin; designed integral-based formulations for chemical reaction dynamics and collected in-flight environmental data using laser-absorption sensors onboard a DC-8 aircraft.

Selected Publications

- **Piland, J.**, Sweet, C., Czajka, A. (2026). DiffGradCAM: A Class Activation Map Using the Full Model Decision to Solve Unaddressed Adversarial Attacks. *Findings of CVPR*.
- **Piland, J.**, Dowling, B., Sweet, C., Czajka, A. (2026). Generalist Multimodal LLMs Gain Biometric Expertise via Human Salience. *IEEE Access*.
- **Piland, J.**, Sweet, C., Czajka, A. (Under review). Divisive Decisions: Utilizing both CAMs in Salience-Based Training to improve Generalization in Binary Classification Tasks. *CVPR Workshop on Biometrics*.
- **Piland, J.**, Sweet, C., Czajka, A. (2025). Multi-level entropy guidance improves model performance in synthetic face detection. *ICIP*.
- **Piland, J.**, Sweet, C., Czajka, A. (2023). Non-generative, energy-based models. *IJCNN*.
- **Piland, J.**, Sweet, C., Czajka, A. (2023). Model focus improves performance of deep learning-based synthetic face detectors. *IEEE Access*.